

AYMANE AIT DADS

Agentic AI Engineering Intern | LLM Orchestration · Tool-Calling · Agent Guardrails

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PROFESSIONAL SUMMARY

Built and deployed an end-to-end agentic AI pipeline integrating Claude API tool-calling, MCP server orchestration, RAG retrieval, and LangChain agent loops to automate multi-step workflows within defined capability boundaries. Engineered a production data science pipeline over 100K+ fiber-optic network records at Orange Maroc, reducing manual analysis time by ~60% through automated feature engineering, clustering, and ensemble modeling. Brings direct experience in LLM orchestration frameworks, prompt engineering with chain-of-thought patterns, vector search and retrieval-augmented generation, and Python-based evaluation harness construction. Available for a Summer 2025 internship; currently based in France and open to USA/Canada placement pending visa sponsorship.

CORE COMPETENCIES

LLM Orchestration | Tool-Calling Protocols | Prompt Engineering | Agent Guardrails | Retrieval-Augmented Generation | Chain-of-Thought Patterns | Evaluation Frameworks | Multi-Agent Workflows

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and shipped an **end-to-end automated pipeline** over 100K+ fiber-optic network records using Random Forest and K-Means, reducing manual analysis time by ~60% and directly informing network optimization decisions.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, enabling the network optimization team to act on model outputs without manual data transformation.
- Implemented automated feature engineering and cross-validated model evaluation, translating raw upload speed, latency, and jitter signals into **interpretable maintenance recommendations** for non-technical stakeholders.
- Delivered stakeholder presentations backed by **Power BI dashboards**, communicating model outputs and business impact across engineering and commercial teams using Python, Scikit-learn, and Pandas.

PROJECTS

Agentic AI Assistant - Claude API + MCP + RAG

Personal Project · 2025 · Production Agentic System

- Architected a **multi-agent orchestration system** using Claude API, LangChain, and MCP server tool-calling to route user intent to the correct agent capability with defined guardrails and escalation logic.
- Built RAG retrieval layer with vector search over a Supabase PostgreSQL backend, enabling agents to self-assess plan confidence and determine when to escalate versus proceed using prompt engineering and few-shot chain-of-thought patterns.

Claude API · LangChain · RAG · Supabase · REST APIs

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress June 2026 · Top 10% · \$106K+ Prize Pool

- Achieved **top 10% public leaderboard (0.80/1.0)** on a \$106K+ Kaggle competition fine-tuning a 30B Mamba-Transformer MoE model using LoRA ($r=32$) with ~888M trainable parameters on 7,828 chain-of-thought reasoning examples via SFTTrainer.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patched allocator warmup, stubbed Mamba/Cutlass imports, and PyTorch RMSNorm fallback - and ran 20+ single-variable ablations confirming completion-only loss and full-linear targeting as critical to the 0.80 score.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Web Applications, Image Security, Statistics

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

AI/LLM & Agentic Systems: LLM Orchestration | Tool-Calling | RAG | LangChain | Prompt Engineering | Claude API | MCP Server

ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | Scikit-learn | PEFT | Unsloth | TRL | LoRA/QLoRA | vLLM

Data & Infrastructure: Python | SQL | Pandas | NumPy | Supabase | PostgreSQL | Docker | Git | Power BI